

VERIFAI

AI-Generated Content Detection Platform

FULL PRODUCT SPECIFICATION DOCUMENT

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1. Executive Summary

VerifAI is a multi-signal forensic detection platform that determines whether images and videos were generated by artificial intelligence. It is designed to serve three user groups: everyday consumers who need quick answers, businesses that require API-scale detection, and legal and forensic professionals who need court-admissible evidence reports.

The platform uses a layered ensemble of six independent detection signals — frequency domain analysis, deep neural classification, noise fingerprinting, metadata forensics, facial anatomy analysis, and temporal consistency checking for video — combined into a weighted confidence score with full explainability output.

Core Ambition

To become the internationally recognised standard for AI-generated content detection, accepted by courts, news organisations, government agencies, and social platforms as the definitive forensic tool for determining the origin of digital media.

2. Product Vision & Market Opportunity

2.1 The Problem

The proliferation of AI image and video generation tools — Midjourney, Stable Diffusion, Flux, Sora, Runway, Kling, Pika, and dozens more — has made it trivially easy to create photorealistic synthetic media. This creates serious problems across multiple domains:

- Misinformation and fake news media using AI-generated imagery
- Legal disputes over whether submitted evidence is authentic
- Insurance fraud using AI-generated images of damage
- Identity fraud using synthetic faces and deepfake videos
- Academic and journalistic integrity failures
- Political manipulation via synthetic media of public figures

2.2 Market Size

Segment	2025 Value	2030 Projected	Growth Rate
AI detection software	\$1.4B	\$8.2B	42% CAGR
Digital forensics market	\$6.9B	\$14.1B	15% CAGR
Content moderation tools	\$4.2B	\$12.8B	25% CAGR
Total addressable market	\$12.5B	\$35.1B	23% CAGR

2.3 Competitive Gaps

Current solutions all have significant weaknesses that VerifAI is designed to address:

- No unified image + video + audio detection in one platform
- No court-admissible forensic report generation with chain of custody
- Poor accuracy on the latest generation models (Flux, Sora, Kling)
- No model-source attribution (which AI tool made this?)
- No continuous retraining pipeline to stay current with new models

3. Full Technical Architecture

3.1 System Overview

VerifAI is built as a cloud-native microservices architecture with a clear separation between the user-facing product layer, the API layer, and the detection engine layer. This separation allows each component to be scaled independently — the detection engine can be given additional GPU resources without touching the frontend.

3.2 Component Map

Layer	Components	Technology
Frontend	Web app, dashboard, report viewer	Next.js 14, TailwindCSS, React
API Gateway	Auth, rate limiting, routing	FastAPI, Redis, JWT
Job Queue	Async task management	Celery + Redis + RabbitMQ
Detection Engine	6-signal analysis pipeline	Python, PyTorch, OpenCV
ML Model Store	Versioned model artifacts	MLflow + S3
Database	Users, scans, results	PostgreSQL + Redis cache
File Storage	Uploads, outputs, reports	AWS S3 + CloudFront CDN
GPU Inference	Model serving	AWS g5.xlarge + ONNX Runtime
Monitoring	Metrics, errors, analytics	Prometheus, Grafana, Sentry

3.3 Full Tech Stack

Frontend

- Next.js 14 with App Router — server-side rendering for SEO, fast page loads
- TailwindCSS — utility-first styling
- Framer Motion — smooth upload and result animations
- Uppy.js — drag-and-drop file upload with chunked transfer for large video files
- D3.js — artifact heatmap visualisation and confidence score charts
- ShadCN UI — accessible component library

Backend / API

- FastAPI (Python) — async API framework, best ML ecosystem integration
- Celery — distributed task queue for processing jobs
- Redis — job queue broker, result caching, session storage
- PostgreSQL — primary relational database for users, scans, reports
- SQLAlchemy — ORM with Alembic for migrations

- Pydantic — data validation and serialisation

ML / Detection Engine

- PyTorch 2.x — primary deep learning framework
- Hugging Face Transformers — Vision Transformer (ViT) models
- TorchVision — image preprocessing and EfficientNet models
- OpenCV — video frame extraction, image preprocessing
- NumPy / SciPy — FFT frequency analysis, signal processing
- Scikit-learn — ensemble weighting, calibration
- ONNX Runtime — optimised model inference (2-4x faster than raw PyTorch)
- FAISS — similarity search for noise fingerprint matching
- Albumentations — data augmentation during training

Infrastructure

- AWS EC2 g5.xlarge — GPU inference instances (NVIDIA A10G)
- AWS S3 — file storage for uploads, models, reports
- AWS CloudFront — CDN for static assets and processed media
- AWS Lambda — lightweight preprocessing, webhook delivery
- AWS SQS — dead-letter queue for failed jobs
- Terraform — infrastructure as code
- Docker + Kubernetes (EKS) — containerised microservices
- GitHub Actions — CI/CD pipeline

4. ML Algorithm Stack — Deep Dive

4.1 What is an Algorithm (For This Context)

An algorithm is a precise, repeatable set of instructions that takes input data and produces a defined output. In VerifAI, each detection signal is its own algorithm. The final verdict is produced by an ensemble algorithm that combines all signal outputs into one weighted score.

Think of it as a panel of six expert witnesses. Each expert examines the same piece of evidence from their specialist angle, gives a confidence score, and the judge (ensemble layer) weighs all opinions to reach a final verdict.

4.2 Signal 1 — Frequency Domain Analysis

Every real photograph has a specific statistical distribution of frequency components, determined by the physics of camera lenses and sensor noise. AI-generated images, even perfect-looking ones, have measurable differences in their frequency spectrum.

How it works

1. Convert image to greyscale
2. Apply 2D Fast Fourier Transform (FFT) to convert from pixel space to frequency space
3. Compute the power spectral density (PSD) — how much energy exists at each frequency
4. Extract statistical features: spectral entropy, high-frequency energy ratio, periodic peak detection
5. Compare extracted signature to trained distributions of real vs AI images

AI diffusion models introduce characteristic high-frequency artifacts at specific spatial frequencies. GANs leave periodic grid patterns detectable in the frequency domain. These are invisible to the human eye but clearly present in the FFT output.

Accuracy Contribution

Frequency analysis alone achieves approximately 78-82% accuracy. It is strongest on diffusion models (Midjourney, Stable Diffusion, DALL-E) and weaker on well post-processed outputs. It contributes 18% weight to the ensemble score.

4.3 Signal 2 — Deep Neural Classifier (CNN + ViT Ensemble)

This is the core classification engine. It uses two neural network architectures working in parallel — a Convolutional Neural Network (CNN) for local texture features and a Vision Transformer (ViT) for global pattern relationships — and combines their outputs.

EfficientNet-B4 (CNN Component)

- Pre-trained on ImageNet, fine-tuned on 4.2M real vs AI image dataset
- Specialises in detecting local texture artifacts — unnatural skin pores, hair strand patterns, background bokeh inconsistencies
- Processes images at multiple scales to catch artifacts at different resolutions
- Outputs a probability vector across 15 generator classes plus 'real' class

Vision Transformer ViT-L/16 (Global Component)

- Divides image into 16x16 patches and models relationships between all patches
- Catches global inconsistencies — lighting that does not match across regions, shadows that point in conflicting directions
- More computationally expensive but catches what CNNs miss
- Attention maps directly feed the explainability heatmap

Combination Strategy

- Both models run in parallel on the same input
- Output logits are concatenated and passed through a lightweight fusion MLP
- Final output: 16-class probability distribution (15 generator families + real)

Accuracy Contribution

The CNN+ViT ensemble alone achieves 89-93% accuracy and contributes 35% weight to the final ensemble. It is the single highest-weight signal. Regular retraining on new generator outputs is what keeps this signal current.

4.4 Signal 3 — Noise Fingerprinting

Every AI model has a unique statistical noise signature — a subtle pattern left in the pixel residuals after the image content is removed. This is analogous to a manufacturer's serial number embedded in the output.

Extraction Process

6. Apply a denoising filter (BM3D or DnCNN) to remove the image content, leaving only the noise residual
7. Extract statistical moments of the residual: mean, variance, skewness, kurtosis
8. Compute the noise autocorrelation pattern
9. Compare against a database of known generator noise profiles using FAISS similarity search
10. Output: matched generator identity + confidence, or 'unknown AI' or 'camera noise'

Real camera images have noise patterns determined by sensor hardware — they are random, isotropic, and follow specific statistical distributions. AI models have deterministic generation processes that leave correlated, non-random noise patterns.

Accuracy Contribution

Noise fingerprinting achieves 85-90% accuracy on known generators and contributes 20% ensemble weight. Its unique value is model attribution — it can often identify which specific AI tool created an image, which is critical for forensic reports.

4.5 Signal 4 — Metadata Forensics

This signal requires no machine learning. It extracts and analyses all metadata embedded in the file and applies a rule-based scoring system.

Metadata Check	Real Image Pattern	AI Image Pattern	Weight
EXIF camera model	Always present (Canon, Nikon, etc.)	Absent or generic	High
Lens data	Focal length, aperture present	Absent	High
GPS coordinates	Often present	Absent	Medium
Creation software	Lightroom, Photos, etc.	Photoshop or absent	Medium
C2PA credentials	May be present	Absent (currently)	High
Thumbnail consistency	Matches main image	May mismatch	Medium
Colour profile	Camera-specific ICC	sRGB generic	Low
Noise model tag	Camera-specific	Absent	Medium

Accuracy Contribution

Metadata forensics alone is only 65-70% accurate (AI tools are adding fake metadata) but it contributes crucial supporting evidence for court reports. It contributes 12% ensemble weight. Its value is explainability — juries understand 'no camera was recorded' more easily than spectral frequency patterns.

4.6 Signal 5 — Facial and Anatomy Analysis

When a human face is present in the image, this specialised module runs additional checks targeting the specific weaknesses of current AI generators.

Checks Performed

- Eye reflection consistency — corneal light reflections should be identical between eyes
- Bilateral symmetry analysis — natural faces have slight asymmetry; AI faces are often too symmetric
- Skin texture frequency analysis — real skin has fractal-like irregularity at multiple scales
- Iris pattern coherence — iris detail should be unique and non-repeating
- Teeth geometry — AI still struggles with realistic, irregular tooth geometry
- Hand anatomy — finger count, proportions, and joint geometry
- Hair strand coherence — AI hair often has repeating strand patterns

Accuracy Contribution

Facial analysis contributes 10% ensemble weight. It is the signal most likely to produce false positives on heavily edited real photos (beauty filters, retouching). It is weighted lower but provides strong supporting evidence when it fires.

4.7 Signal 6 — Temporal Consistency (Video Only)

For video content, this signal analyses consistency across frames — something that is trivially maintained in real footage but extremely difficult for AI generators to achieve perfectly.

- Object permanence tracking — do objects maintain consistent shape, colour, and size across frames?
- Optical flow anomalies — AI video often has unnatural motion vectors at object boundaries
- Texture flickering — AI-generated textures subtly change between frames
- Physics violations — hair, fabric, water behave unnaturally in AI video
- Background coherence — AI backgrounds often flicker or morph between frames
- Lighting consistency — shadows and highlights should not shift randomly between frames

4.8 The Ensemble Algorithm

The ensemble layer combines all six signal outputs into one final verdict. It is not a simple average — it is a calibrated weighted combination trained to be more accurate than any individual signal.

Signal	Base Weight	Boosted When	Reduced When
Neural classifier (CNN+ViT)	35%	High resolution image	Heavy compression
Noise fingerprinting	20%	Low JPEG compression	Small file size
Frequency analysis (FFT)	18%	PNG / lossless format	Multiple re-saves
Metadata forensics	12%	Original upload	Screenshot of AI image

Facial analysis	10%	Face detected in frame	No face present
Temporal consistency	5% (image) / 25% (video)	Video input	Single image

Weights are dynamically adjusted per scan based on file type, compression level, resolution, and face presence. The ensemble also produces an uncertainty score — if signals strongly disagree, the system reports 'uncertain' rather than forcing a verdict.

5. Video Processing Architecture & Performance

5.1 The Video Problem

Video is fundamentally harder to process than images for three reasons: file size can reach gigabytes, processing a video frame-by-frame is massively expensive, and temporal analysis requires keeping multiple frames in memory simultaneously. Naive approaches either run out of memory, take minutes per scan, or produce inaccurate results.

5.2 The Solution: Intelligent Frame Sampling

The single most important optimisation is not processing every frame. A 60-second video at 30fps has 1,800 frames. Processing all of them would take 3-8 minutes per video. Instead, VerifAI uses an intelligent sampling strategy:

Adaptive Keyframe Extraction

11. Extract frames using scene change detection (not uniform time intervals)
12. Identify high-motion frames vs low-motion frames using optical flow magnitude
13. Sample 1 frame per second from low-motion segments
14. Sample 3 frames per second from high-motion segments (where AI artifacts are most visible)
15. Always include first and last 5 seconds at full sample rate (AI generators show most artifacts at start and end)

Result: A 60-second video produces approximately 80-120 representative frames instead of 1,800. This reduces compute by over 93% with less than 2% accuracy loss.

5.3 Parallel Processing Pipeline

Sampled frames are not processed sequentially — they are batched and processed in parallel across multiple GPU threads:

Frame-Level Parallelism

- Frames are batched into groups of 32 (optimal for GPU memory)
- Each batch is processed simultaneously on the GPU
- CNN and ViT run in parallel on the same batch
- For a 90-frame sample, this means 3 batches instead of 90 sequential passes

Signal Parallelism

- Frequency analysis, noise fingerprinting, and metadata extraction run concurrently on separate CPU workers
- Neural classifier runs on GPU while other signals run on CPU — no waiting

- Results are collected asynchronously and merged at the ensemble layer

5.4 Video-Specific Optimisations

Problem	Naive Approach	VerifAI Solution	Speedup
Large video upload	Wait for full upload	Chunked upload + stream processing	5-10x
Frame extraction	Decode all frames	GPU-accelerated decode (NVDEC)	8x
Processing all frames	Sequential frame-by-frame	Adaptive sampling + batch GPU	15-20x
High-res 4K video	Downsample first	Multi-scale pyramid analysis	3x
Multiple simultaneous uploads	Queue and wait	Horizontal GPU worker scaling	Linear
Memory overflow on long video	Load full video to RAM	Sliding window with frame eviction	Infinite scalable

5.5 Client-Side Pre-Processing (Reducing Server Load)

Before the file even reaches the server, the browser does light work to reduce what is sent:

- Generate a cryptographic SHA-256 hash of the file in the browser (proves file integrity, enables deduplication)
- Extract basic metadata client-side (no server round-trip needed for metadata forensics input)
- For images over 20MB, generate a lossless-compressed copy for transmission while keeping the original hash for verification
- For video, extract the first 5 seconds locally and send them first — server starts analysis immediately while the rest of the file uploads

5.6 Caching and Deduplication

The single biggest server load reduction is simply not processing the same file twice:

- Every uploaded file is SHA-256 hashed before upload begins
- Hash is checked against a database of previously scanned files
- If a match exists, the cached result is returned instantly — zero GPU time used
- This eliminates the most common pattern: the same viral AI image being uploaded thousands of times by different users
- Expected cache hit rate at scale: 40-60% of all uploads

5.7 Target Performance Benchmarks

File Type	File Size	Target Analysis Time	Method
Image (JPG/PNG)	Under 5MB	Under 8 seconds	Full 6-signal pipeline
Image (JPG/PNG)	5-20MB	8-15 seconds	Full pipeline + downscale copy
Video (MP4)	Under 100MB	Under 25 seconds	Adaptive sampling + parallel
Video (MP4)	100MB-500MB	25-60 seconds	Chunked + streaming analysis
Video (MP4)	500MB-2GB	60-180 seconds	Multi-worker + priority queue
4K Video	Any size	Add 30% to above	Multi-scale pyramid

6. Development Phases

Phase 1 — Foundation (Months 1-3)

Goal: Working MVP with image detection. Target accuracy: 85%.

- Core web application with upload interface
- Metadata forensics module
- Frequency analysis (FFT) module
- Basic CNN classifier (fine-tuned EfficientNet-B4)
- Simple confidence score output with 3 signal breakdown
- User accounts and scan history
- Basic API endpoint for developers

Phase 2 — Depth (Months 4-6)

Goal: High-accuracy image detection with full explainability. Target accuracy: 92%.

- Vision Transformer integration and CNN+ViT ensemble
- Noise fingerprinting module with initial generator database (10 models)
- Facial and anatomy analysis module
- Heatmap visualisation output
- Full ensemble scoring with dynamic weighting
- Explainability report generator (plain English summaries)
- API v2 with webhook support

Phase 3 — Video (Months 7-10)

Goal: Full video detection capability. Target accuracy: 88% on video.

- Adaptive keyframe sampling pipeline
- Temporal consistency analysis module
- GPU-accelerated video decode (NVDEC integration)
- Chunked upload support for files up to 2GB
- Parallel processing worker pool
- Client-side pre-processing (hash, metadata, first-frame preview)

Phase 4 — Legal Grade (Months 11-15)

Goal: Court-admissible forensic reports and industry standard positioning.

- Chain of custody system (cryptographic file hashing, tamper-proof audit logs)
- Signed forensic report generation (PDF with digital certificate)
- Third-party validation partnerships (university research labs)

- NIST evaluation submission
- Expert witness programme — certified analyst network
- C2PA content credential verification

Phase 5 — Scale & Moat (Months 16-24)

Goal: Enterprise-grade product with continuous learning pipeline.

- Automated retraining pipeline — detects new AI model releases and retrains within 30 days
- Generator fingerprint database expanded to 50+ models
- Browser extension for in-context scanning
- Enterprise SSO, SLA, and dedicated support
- White-label API for platform partners
- Mobile apps (iOS and Android)

7. Business Model & Monetisation

7.1 Revenue Streams

Plan	Price	Includes	Target User
Free	\$0/month	5 scans/month, basic verdict	Curious consumers
Pro	\$29/month	500 scans, full signal breakdown, heatmap	Journalists, researchers
Business	\$99/month	Unlimited scans, API access, bulk upload	SMBs, agencies
Enterprise	Custom	SLA, white-label, dedicated GPU, support	Platforms, media cos
Forensic Report	\$150-500/report	Signed PDF, chain of custody, court-ready	Legal professionals
API (pay-per-use)	\$0.08/image, \$0.40/video	Developer integration	Developers, platforms

7.2 Revenue Projections

Year	Paying Users	API Calls/month	Monthly Revenue	Annual Revenue
Year 1	500	50,000	\$18,000	\$216,000
Year 2	5,000	500,000	\$165,000	\$1.98M
Year 3	20,000 + enterprise	5M+	\$580,000	\$6.96M

8. Legal & Forensic Grade Standards

8.1 What Courts Require

For evidence to be accepted in court, it must meet the Daubert standard (US) or equivalent — the methodology must be scientifically valid, peer-reviewed, tested with known error rates, and widely accepted in the relevant field.

Requirements VerifAI Must Meet

- Published, peer-reviewed methodology
- Independently validated accuracy with known error rates
- Reproducibility — same input always produces same output
- Chain of custody — provable that the analysed file was not tampered with

- Qualified human expert who can testify to the methodology
- Daubert hearing readiness — prepared responses to methodological challenges

8.2 Chain of Custody System

16. File is hashed (SHA-256) in the browser before upload begins
17. Hash is recorded with a UTC timestamp the moment the file reaches the server
18. File is stored in write-once S3 bucket with versioning enabled
19. All analysis steps are logged with timestamps in a tamper-evident audit trail
20. Final report is signed with a PKI certificate tied to VerifAI's identity
21. Hash of the report itself is also recorded — proving the report was not edited after generation

8.3 Forensic Report Contents

- Case reference number and submission timestamp
- SHA-256 hash of analysed file
- System version and model version used for analysis
- Full signal breakdown with per-signal confidence scores
- Heatmap showing artifact locations
- Plain English summary written for non-technical readers
- Model attribution section (which AI generator most likely produced this)
- Digital signature and certificate chain
- Analyst certification block (space for human expert countersignature)

9. Risk Analysis

Risk	Likelihood	Impact	Mitigation
New AI models evade detection	High	High	30-day retraining pipeline, continuous monitoring
False positives on real images	Medium	High	Uncertainty threshold, human review flag
Legal challenge to methodology	Medium	High	Peer review, third-party validation, Daubert prep
Competitor copies approach	High	Medium	Proprietary dataset, speed of iteration, legal moat
GPU cost exceeds revenue	Low	High	Aggressive caching, ONNX optimisation, spot instances
Regulatory change on AI detection	Low	Medium	Legal counsel, policy engagement, EU AI Act compliance

10. Team & Resource Requirements

Phase 1-2 Team (6 people)

- 1 x ML Engineer — model training, signal pipeline
- 1 x Backend Engineer — FastAPI, Celery, database
- 1 x Frontend Engineer — Next.js, upload flow, results UI
- 1 x DevOps / Infrastructure — AWS, Kubernetes, CI/CD
- 1 x Product / Founder — roadmap, partnerships, legal strategy
- 1 x Data Annotator / Researcher — building and labelling training dataset

Phase 3-5 Additional Hires

- 1 x Video ML Specialist — temporal analysis, video pipeline
- 1 x Forensic Consultant — legal grade standards, court readiness
- 1 x Enterprise Sales — platform partnerships, API deals
- 2 x ML Engineers — model retraining, signal expansion

Infrastructure Costs (Monthly)

Resource	Phase 1	Phase 2-3	Phase 4-5
GPU instances (inference)	\$800	\$3,200	\$12,000+

S3 storage + CDN	\$120	\$480	\$2,000+
Database (RDS + Redis)	\$200	\$600	\$2,400+
Misc (monitoring, CI/CD)	\$180	\$320	\$600+
Total infrastructure	\$1,300	\$4,600	\$17,000+

VERIFAI

Deep Technical & Business Strategy Document

What Makes Us the Global Standard | Benchmarks | Retraining Pipeline | Database Schema | Funding Strategy

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1. What Makes VerifAI the Industry Standard

Being accurate is not enough to become the global standard. Accuracy gets you into the market. What makes you the standard is a combination of scientific credibility, speed, cost efficiency, explainability, and continuous improvement. Here is every element that needs to be true from day one.

1.1 The Five Pillars of Standard Status

Pillar 1 — Scientific credibility

- Methodology published as peer-reviewed white papers before launch
- Independent accuracy tests by at least two universities (target: MIT Media Lab, Oxford Internet Institute)
- Known, published error rates — not hidden. Transparency builds trust
- Submitted to NIST (US National Institute of Standards and Technology) for evaluation
- Methodology documented well enough that a PhD student could replicate it

Pillar 2 — Accuracy that no competitor can match

- Target 94%+ on images, 90%+ on video at launch
- Ensemble of 6 signals — no single point of failure
- Model-specific attribution (we do not just say AI — we say which AI tool)
- Uncertainty output — when we are not sure, we say so. Courts trust honesty more than false confidence
- Retraining pipeline that matches any new AI model release within 30 days

Pillar 3 — Explainability that courts and journalists accept

- Every verdict comes with plain-English reasons, not just a score
- Heatmap showing exactly where in the image/video the artifacts are
- Per-signal breakdown so a human expert can explain each component in court
- Reproducibility guarantee — same file always produces same output
- Signed, timestamped, chain-of-custody reports

Pillar 4 — Speed and cost that dominate at scale

- Image result in under 8 seconds (competitors average 20-40 seconds)
- Video result in under 30 seconds for files under 100MB
- Cost per scan under \$0.004 at scale (CPU+GPU combined)
- Cache deduplication reduces GPU usage by 40-60% at scale
- ONNX inference means 3x faster than raw PyTorch with same accuracy

Pillar 5 — Continuous moat

- Proprietary dataset that grows with every user scan (with consent)
- Automated retraining triggers when new AI models are detected in the wild
- Generator fingerprint database that no competitor can easily replicate
- API partnerships that make VerifAI the embedded standard inside other platforms

The Single Most Important Thing

Our moat is not our initial model — it is our retraining pipeline and our proprietary labeled dataset. Every scan that comes through VerifAI (with user consent) adds to a dataset that trains our next model. After 12 months we will have millions of labeled examples across every major generator. That dataset is worth more than the model itself and cannot be bought or replicated quickly.

1.2 Revised Development Roadmap — Standard-Focused

Phase	Duration	Standard-Building Milestone	Accuracy Target
0 — Foundation	Months 1-2	Open-source the benchmark test suite. Invite researchers to validate.	N/A — build only
1 — MVP	Months 3-5	Launch with published white paper. Metadata + FFT + basic CNN.	85% images
2 — Precision	Months 6-9	CNN+ViT ensemble. Noise fingerprinting. First university validation.	92% images
3 — Video	Months 10-14	Full video pipeline. Temporal analysis. 30-day retraining cycle.	90% video, 94% image
4 — Legal grade	Months 15-18	NIST submission. Forensic report. Chain of custody. Expert witness network.	Court accepted
5 — Standard	Months 19-24	Platform API partnerships. C2PA integration. Browser extension. Election bodies.	Industry standard

2. Benchmarks We Must Hit to Be the Standard

These are not aspirational numbers. These are the specific benchmarks, measured in specific ways, that will determine whether VerifAI is taken seriously as a standard or dismissed as just another tool.

2.1 Accuracy Benchmarks

Metric	Phase 1 Target	Phase 2 Target	Phase 3 Target	Industry Standard Level
Image detection (overall)	85%	92%	94%+	95%+
Image — diffusion models	82%	91%	94%	95%+
Image — GAN models	88%	93%	95%	96%+
Image — post-processed AI	72%	82%	87%	90%+
Video detection (overall)	N/A	N/A	90%	92%+
Video — temporal artifacts	N/A	N/A	88%	91%+
False positive rate (real = AI)	<8%	<5%	<3%	<2%
False negative rate (AI = real)	<18%	<10%	<7%	<5%
Model attribution accuracy	N/A	78%	88%	90%+
Unknown model detection	N/A	70%	80%	85%+

How to Measure These

Every quarter, run the full test suite against a held-out benchmark dataset of 50,000 real images and 50,000 AI images across all major generators. The dataset must be updated monthly to include outputs from new model releases. Publish all results publicly — this is what makes you trusted.

2.2 Performance Benchmarks

Metric	Target	How to Measure	Acceptable Minimum
Image analysis time (p50)	Under 6 seconds	Median of 10,000 real scans	Under 10 seconds
Image analysis time (p99)	Under 15 seconds	99th percentile of scans	Under 25 seconds
Video analysis time (100MB, p50)	Under 25 seconds	Median of 1,000 video scans	Under 45 seconds
API response time	Under 200ms to job acceptance	Time from POST to 202 response	Under 500ms

System uptime	99.9% monthly	Prometheus uptime monitoring	99.5%
Concurrent scan capacity	500 simultaneous scans	Load test with k6	200 simultaneous
Cache hit rate at scale	40%+	Redis hit/miss ratio	25%
Cost per image scan	Under \$0.008	AWS cost / scan count monthly	Under \$0.02
Cost per video scan	Under \$0.05	AWS GPU cost / scan count	Under \$0.12

2.3 Legal Grade Benchmarks

Requirement	Target	Validation Method
Reproducibility	100% — same file = same result	Automated test: scan same file 100x, verify identical output
Report signing	PKI certificate on every report	SSL/TLS certificate chain verification
Chain of custody	SHA-256 hash logged at upload	Audit log integrity check
White paper published	Before Phase 2 launch	Public URL, DOI assigned
Third-party validation	2 universities by Phase 4	Published validation study
NIST evaluation	Submitted by Phase 4	NIST PFT or FATE programme
Expert witness qualified	3 certified analysts by Phase 4	Court qualification hearing

3. Tools and Systems to Build for Top-Notch Detection

3.1 The Core Detection Tools Stack

Component	Tool / Library	Why This One	Alternative
Deep learning framework	PyTorch 2.x	Best research community, most AI detection research uses it	TensorFlow (slower iteration)
Vision Transformer	timm library (ViT-L/16)	Largest pretrained ViT models, easy fine-tuning	HuggingFace ViT
CNN backbone	EfficientNet-B4 via timm	Best accuracy/compute tradeoff for image classification	ResNet-50 (lower accuracy)
Inference optimisation	ONNX Runtime + TensorRT	3-4x speedup over PyTorch, no accuracy loss	OpenVINO (Intel only)
Frequency analysis	NumPy FFT + PyWavelets	Gold standard for spectral analysis, well tested	SciPy (similar)
Noise fingerprinting	FAISS (Facebook AI)	Billion-scale similarity search, runs on GPU	Annoy (CPU only)
Video decode	PyAV + NVDEC (GPU)	8x faster than CPU decode for H.264/H.265	OpenCV (CPU only)
Optical flow	RAFT model (PyTorch)	State of art optical flow, detects AI motion artifacts	Farneback (less accurate)
Face detection	InsightFace	Best open-source face analysis, handles partial faces	MediaPipe (less accurate)
Metadata extraction	ExifTool + pyexiv2	Most complete EXIF/XMP/IPTC support available	Pillow (incomplete)
Model experiment tracking	MLflow	Tracks every training run, model versions, metrics	W&B (paid at scale)
Data augmentation	Albumentations	Fastest image augmentation library, GPU support	torchvision transforms

3.2 Proprietary Systems We Must Build Ourselves

These are the systems that cannot be bought off the shelf. They are the source of our competitive moat and they must be built from scratch.

System 1 — The Benchmark Dataset Pipeline

A continuously updated labeled dataset of real and AI-generated content. This is our most valuable asset.

- Automated scraper that collects verified real photos from Flickr, Unsplash, and camera RAW databases (with licenses)
- Automated pipeline that generates synthetic images from every major AI model on a weekly schedule
- Human annotation queue for borderline cases (real photographers reviewing outputs)
- Version-controlled dataset with full provenance — every image tracked to its source
- Target dataset size: 500K labeled pairs at launch, 5M by end of Year 2

System 2 — The Generator Fingerprint Database

A database of noise signatures and statistical fingerprints for every known AI model family.

- One entry per model version (Midjourney v5, v6, Flux 1.0, Flux 1.1, Sora, etc.)
- Each entry contains: noise PSD signature, frequency domain template, color distribution profile
- FAISS index over all entries for sub-millisecond similarity search
- Automated update trigger: when a new model is detected in the wild, pipeline generates 1,000 samples and adds a new fingerprint entry
- Target: 15 model families at launch, 50+ by end of Year 2

System 3 — The New Model Early Warning System

An automated system that detects when a new AI image or video generator has been released.

- Monitor AI model release blogs, HuggingFace new model submissions, Reddit r/StableDiffusion, Twitter/X AI accounts
- When a new model is mentioned, automatically generate test images and run them through the current detector
- If detection accuracy drops below 75% on new model outputs, trigger a retraining alert
- Human-in-the-loop review before retraining begins
- Target: detect and respond to new major models within 2 weeks of release

System 4 — The Continuous Retraining Pipeline

This is covered in full in Section 4. It is the most important system we build.

System 5 — The Explainability Engine

A system that converts raw model outputs into human-readable, court-ready explanations.

- GradCAM and attention map extraction to generate pixel-level heatmaps
- Natural language template system: takes signal scores and produces plain English paragraphs
- Legal language review — templates reviewed by forensic linguist before launch
- Jurisdiction-aware output: slightly different language for US courts vs UK courts vs EU courts

- Confidence calibration: language changes based on certainty level (definitive vs probable vs uncertain)

4. The Continuous Retraining Pipeline

This is the single most important system in VerifAI. Any detection model trained once will degrade over time as AI generators improve. The retraining pipeline is what keeps VerifAI accurate permanently, and it is what creates an insurmountable moat — because our dataset grows with every scan.

4.1 Why Retraining is the Moat

The Arms Race Reality

When Midjourney releases v7, our v6-trained model will lose accuracy on those outputs. A competitor who retrains once per quarter will always be 3 months behind new models. Our 30-day retraining cycle means we are always current. After 2 years, our proprietary dataset will contain millions of labeled examples from every generator version ever released. That dataset cannot be quickly replicated by a new entrant — it took 2 years to build.

4.2 The Full Retraining Pipeline Architecture

Trigger Layer — What starts a retraining run

1. Scheduled trigger: every 30 days automatically, regardless of model changes
2. Performance trigger: if accuracy on the benchmark test suite drops below 91%, immediate trigger
3. New model trigger: early warning system detects a new generator (see Section 3, System 3)
4. Data volume trigger: when the training dataset has grown by 100,000 new labeled examples

Data Collection Layer — Building the training batch

5. Pull all new user-consented scans since last training run
6. Automatically generate 5,000 synthetic images from each known AI model family
7. Pull 5,000 verified real images from the licensed real photo pipeline
8. Run all new samples through the current model and flag low-confidence predictions for human review
9. Human annotators review flagged samples (target: 200 samples per day per annotator)
10. Merge new data with existing dataset, deduplicate by hash

Training Layer — Running the new model

11. Fine-tune EfficientNet-B4 and ViT-L/16 on updated dataset for 10 epochs
12. Use mixed-precision training (FP16) to reduce GPU memory and training time by 40%
13. Evaluate on held-out benchmark dataset after each epoch

14. Early stopping if validation accuracy plateaus for 3 consecutive epochs
15. Total training time target: under 6 hours on 4x A100 GPUs
16. Log all metrics to MLflow: loss curves, per-generator accuracy, confusion matrix

Validation Layer — Approving the new model

17. New model must score higher than previous model on the standard benchmark
18. New model must not regress on any single generator family by more than 2%
19. False positive rate must remain below 3%
20. Reproducibility test: 500 identical scans must produce identical results
21. If validation passes, promote to shadow mode (runs alongside production, results logged but not shown)
22. Shadow mode for 48 hours: compare new model output to current model on live traffic
23. If shadow mode results are consistent, promote to production with zero-downtime deployment

Rollback Layer — Safety net

24. Every model version is stored permanently in MLflow + S3
25. One-command rollback to any previous model version
26. Automated rollback trigger: if production accuracy drops below 88% for 1 hour, auto-rollback
27. All forensic reports record which model version was used — historical reports remain valid

4.3 Retraining Infrastructure

Component	Tool	Purpose
Training compute	AWS p4d.24xlarge (8x A100)	Training runs — spot instances to reduce cost by 70%
Experiment tracking	MLflow self-hosted	Every run logged, models versioned, metrics compared
Dataset versioning	DVC (Data Version Control)	Git for datasets — full provenance of every training set
Pipeline orchestration	Apache Airflow	Schedules all pipeline steps, handles failures gracefully
Model registry	MLflow Model Registry	Staging, production, archived model states
Container registry	AWS ECR	Stores Docker images for each model version
Feature store	Feast	Stores precomputed image features to avoid recomputing on retrain
Data labeling	Label Studio (self-hosted)	Human annotation queue for borderline cases

Monitoring	Evidently AI	Detects model drift in production — triggers retraining alert
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5. Database Schema

The database is designed around four core concerns: scan results must be fast to write and query, forensic reports must be immutable once created, user data must be isolated and deletable (GDPR compliance), and the generator fingerprint database must support fast similarity search.

5.1 Primary Database — PostgreSQL

users table

Column	Type	Notes
id	UUID PRIMARY KEY	Auto-generated, never sequential (security)
email	VARCHAR(255) UNIQUE NOT NULL	Lowercase, indexed
password_hash	VARCHAR(255)	bcrypt, never store plaintext
plan	ENUM('free','pro','business','enterprise')	Determines rate limits and features
api_key_hash	VARCHAR(255)	Hashed API key for developer access
created_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	All timestamps UTC
deleted_at	TIMESTAMPTZ NULL	Soft delete for GDPR — data zeroed on deletion
consent_dataset	BOOLEAN DEFAULT FALSE	User consent to use scans for training data
org_id	UUID REFERENCES organisations(id)	For enterprise multi-user accounts

scans table

Column	Type	Notes
id	UUID PRIMARY KEY	Scan identifier, used in report references
user_id	UUID REFERENCES users(id)	Nullable — public API scans have no user
file_hash	VARCHAR(64) NOT NULL INDEXED	SHA-256 of original file — enables deduplication
file_type	ENUM('image','video')	
file_size_bytes	BIGINT	

file_name	VARCHAR(255)	Original filename, sanitised
s3_key	VARCHAR(512)	Storage path in S3 — null if deduplicated
status	ENUM('queued','processing','complete','failed')	Indexed for queue queries
verdict	ENUM('ai','real','uncertain')	Final verdict
confidence_score	DECIMAL(5,2)	0.00 to 100.00
model_attribution	VARCHAR(100)	e.g. midjourney_v6, flux_1_1, unknown_ai
attribution_confidence	DECIMAL(5,2)	Confidence in model attribution specifically
processing_time_ms	INTEGER	Total analysis time in milliseconds
model_version	VARCHAR(50) NOT NULL	Which VerifAI model version ran this scan
created_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	
completed_at	TIMESTAMPTZ	
is_cached	BOOLEAN DEFAULT FALSE	True if result came from cache

signal_results table

One row per signal per scan. Stores the raw output of each of the six detection signals.

Column	Type	Notes
id	UUID PRIMARY KEY	
scan_id	UUID REFERENCES scans(id) ON DELETE CASCADE	
signal_name	ENUM('frequency','neural','noise','metadata','facial','temporal')	
score	DECIMAL(5,2)	0-100 probability of AI for this signal
triggered	BOOLEAN	Did this signal fire positively?
weight_applied	DECIMAL(4,3)	The dynamic weight used in ensemble for this scan
detail_json	JSONB	Signal-specific detail: FFT peaks,

		matched generator, missing EXIF fields, etc.
processing_time_ms	INTEGER	

forensic_reports table

Column	Type	Notes
id	UUID PRIMARY KEY	
scan_id	UUID REFERENCES scans(id)	
user_id	UUID REFERENCES users(id)	
report_hash	VARCHAR(64) NOT NULL	SHA-256 of the report PDF — proves it was not altered
s3_key	VARCHAR(512)	Path to signed PDF in write-once S3 bucket
case_reference	VARCHAR(100)	User-provided case number for legal use
digital_signature	TEXT	PKI signature over report hash
certificate_chain	TEXT	Full certificate chain for signature verification
analyst_id	UUID REFERENCES analysts(id) NULL	If countersigned by a certified analyst
jurisdiction	VARCHAR(50)	us, uk, eu, other — affects report language
created_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	Immutable after creation

generator_fingerprints table

Column	Type	Notes
id	UUID PRIMARY KEY	
generator_name	VARCHAR(100) NOT NULL	e.g. midjourney
generator_version	VARCHAR(50) NOT NULL	e.g. v6_0
noise_signature_vector	VECTOR(512)	pgvector column — enables similarity search
frequency_template	JSONB	FFT signature template as JSON array
color_distribution	JSONB	Statistical moments of color channels
sample_count	INTEGER	How many samples this fingerprint is based on

confidence_threshold	DECIMAL(5,2)	Minimum similarity score to call a match
active	BOOLEAN DEFAULT TRUE	Set false when superseded by new version
created_at	TIMESTAMPTZ	
updated_at	TIMESTAMPTZ	

audit_log table (append-only)

Column	Type	Notes
id	BIGSERIAL PRIMARY KEY	Sequential — not UUID (enables fast append)
event_type	VARCHAR(100) NOT NULL	file_uploaded, scan_started, scan_completed, report_created, etc.
scan_id	UUID	
user_id	UUID	
ip_address	INET	
event_data	JSONB	Event-specific payload
file_hash_at_event	VARCHAR(64)	Proves file was not changed between events
created_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	Index this column — high query volume

5.2 Key Indexes

- scans(file_hash) — enables O(1) deduplication check
- scans(user_id, created_at DESC) — user scan history queries
- scans(status) WHERE status != 'complete' — job queue polling
- audit_log(scan_id) — court report generation
- audit_log(created_at) — time-range queries for compliance
- generator_fingerprints using pgvector IVFFlat index — sub-5ms similarity search

5.3 Cache Layer — Redis

Key Pattern	Value	TTL	Purpose
scan:result:{file_hash}	Full result JSON	30 days	Primary deduplication cache
user:ratelimit:{user_id}	Request count	1 hour rolling	Rate limiting per plan
job:status:{scan_id}	Job status enum	24 hours	Real-time status polling
fingerprint:index	FAISS index blob	Until retrain	In-memory similarity search

model:current_version	Version string	Until deploy	Consistent version across workers
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6. Business Model and Funding Strategy

6.1 Revenue Architecture

VerifAI has three revenue layers designed to work together: consumer and SMB SaaS builds brand recognition and user base, the API layer converts that recognition into recurring B2B revenue, and legal/forensic reports are high-margin one-off transactions that also build credibility.

Revenue Stream	Price	Margin	Volume Driver
Free tier	\$0	Negative (loss leader)	Builds user base, feeds training data
Pro plan	\$29/month	~80%	Individual creators, journalists, researchers
Business plan	\$99/month	~82%	Agencies, media companies, legal firms
Enterprise (annual)	\$2,400-\$24,000/year	~75%	Platform partnerships, government
API pay-per-use	\$0.08/image, \$0.40/video	~70%	Developers, moderation tools
Forensic report	\$150-\$500 each	~85%	Legal cases, insurance claims
White-label API	\$5,000-\$50,000/month	~78%	Social platforms, news agencies
Certification badge	\$200-\$2,000/month	~90%	Publishers, news sites

6.2 Detailed Revenue Projections

Year	Pro Users	Business Users	Enterprise Clients	API Revenue	Forensic Reports	Total ARR
Year 1	300 @ \$29	80 @ \$99	2 @ \$5k	\$8k	\$15k	\$228k
Year 2	2,500 @ \$29	500 @ \$99	15 @ \$8k avg	\$80k	\$120k	\$1.97M
Year 3	8,000 @ \$29	2,000 @ \$99	60 @ \$12k avg	\$400k	\$500k	\$6.5M
Year 4	20,000 @ \$29	6,000 @ \$99	200 @ \$15k avg	\$1.2M	\$1.5M	\$17.8M
Year 5	50,000 @ \$29	15,000 @ \$99	500 @ \$20k avg	\$3M	\$4M	\$46M

6.3 Cost Structure

Cost Category	Year 1	Year 2	Year 3	Notes
Infrastructure (AWS)	\$18k	\$72k	\$240k	GPU costs dominate; ONNX + caching keep this low
Team salaries	\$280k	\$720k	\$1.8M	6 people Y1, grows to 20 by Y3
Data annotation	\$24k	\$60k	\$120k	Human labelers for borderline cases
Legal + compliance	\$15k	\$40k	\$80k	Forensic certification, NIST, court prep
Marketing + sales	\$12k	\$80k	\$300k	Content, conferences, enterprise sales
Research partnerships	\$10k	\$30k	\$60k	University validation, white papers
Total OpEx	\$359k	\$1.002M	\$2.6M	

6.4 Funding Strategy

Phase 0-1 — Bootstrap / Pre-seed (\$0 to \$300k)

Build the MVP without external funding if possible. The goal here is to have a working product with published accuracy numbers before approaching investors. Investors in AI tooling want to see technical credibility, not just a pitch deck.

- Founder capital or friends and family round: \$50-150k to cover infrastructure and initial hires
- Apply for AI/deep tech grants: Nigerian iTDA grants, Tony Elumelu Foundation, Google for Startups Africa (\$200k non-dilutive)
- Enter competitions: MIT Solve, Falling Walls, MIT Media Lab AI Ethics competitions
- Build in public: publish the benchmark methodology before the product launches. Researchers will cite you. Citations are credibility.

Phase 2 — Pre-seed / Seed (\$300k to \$2M)

Once accuracy benchmarks hit 90%+ and you have 50+ paying customers, raise a proper seed round.

- Target: \$1-2M at \$6-10M pre-money valuation
- Investor profile: AI-focused seed funds (Basis Set Ventures, AIX Ventures, Radical Ventures), media-tech angels, cybersecurity VCs
- Pitch angle: the legal-grade forensic standard play. Frame it as infrastructure for trust in digital media, not just another detection tool
- Use of funds: 60% team, 25% infrastructure scale, 15% legal/certification track

Phase 3-4 — Series A (\$2M to \$15M)

Once video detection is live and first enterprise contracts are signed, raise Series A.

- Target: \$8-15M at \$40-60M pre-money
- Investor profile: enterprise SaaS VCs (Bessemer, Sequoia, a16z cultural/media focus), strategic media companies as investors
- Strategic investors to target: AP, Reuters, BBC, major insurance companies (they have strong motive to detect AI fraud)
- Pitch angle: by this point, lead with traction. X enterprise clients, Y API calls per month, Z forensic reports filed in actual legal cases

Non-dilutive Funding Sources (Pursue in Parallel)

- DARPA media forensics programme (US): up to \$5M for AI detection research
- EU Horizon Europe calls on AI transparency and media integrity: €500k-2M grants
- UK UKRI Trusted AI programme: up to £2M
- Partnership R&D with news agencies: AP, Reuters, BBC often pay for exclusive early access to new detection tools
- NIST SBIR grants for small businesses doing forensic standards work

6.5 Path to Profitability

Month	MRR	Monthly Costs	Monthly Profit/Loss	Cumulative
Month 6	\$3k	\$28k	-\$25k	-\$90k
Month 12	\$19k	\$35k	-\$16k	-\$250k
Month 18	\$65k	\$65k	Breakeven	-\$400k
Month 24	\$165k	\$95k	+\$70k	-\$250k
Month 30	\$350k	\$150k	+\$200k	+\$100k
Month 36	\$650k	\$220k	+\$430k	+\$1.1M

The Key Insight on Profitability

Infrastructure costs grow much slower than revenue because of the caching and ONNX optimisations. Once 40%+ of scans are cache hits (free to serve), the marginal cost of each additional user drops sharply. At 20,000 users, we estimate 55% of all scans are cache hits — meaning over half our volume costs essentially nothing to serve.

VERIFAI

Full Development Roadmap, Cost Breakdown & Financial Model

Version 3.0 | June 2026 | Confidential

1. Full Development Roadmap (Month by Month)

This roadmap covers 24 months from zero to industry standard. Each month has specific deliverables, milestones, and the person or team responsible. Nothing is vague — every month has a clear definition of done.

Phase 0 — Pre-Build (Months 1-2)

Goal

Have the scientific foundation, dataset pipeline, and development environment fully set up before writing a single line of product code. Most startups skip this phase and pay for it later. We do not.

Month	Deliverable	Definition of Done	Owner
1	Research review	Read and summarise top 20 academic papers on AI image detection. Extract key signal methods we will use.	ML Engineer
1	Dataset pipeline v1	Automated script collects 10,000 real images (Flickr API, Unsplash API) and generates 10,000 AI images across 5 generators.	ML Engineer
1	Dev environment	Docker Compose setup running FastAPI + PostgreSQL + Redis locally. All engineers can run the stack in one command.	Backend Engineer
1	Benchmark test suite	Python script that takes a model and outputs accuracy, FP rate, FN rate across all generator families. Will be used every sprint.	ML Engineer
2	Frequency analysis module	Standalone Python module: input image, output FFT features + anomaly score. Unit tested. Accuracy 72%+ on test set.	ML Engineer
2	Metadata extraction module	Standalone module: input file, output 8 metadata signals + rule-based score. Works on JPG, PNG, WEBP.	Backend Engineer
2	White paper draft v1	10-page technical paper describing our methodology. Not published yet — internal review only.	ML Engineer + Founder
2	AWS account + Terraform	All infrastructure defined as code. One command deploys a full staging environment.	DevOps

Phase 1 — MVP (Months 3-5)

Goal

A working product that real users can use. Image detection only. Target: 85% accuracy. Launch publicly at end of Month 5.

Month	Deliverable	Definition of Done	Owner
3	CNN classifier v1	EfficientNet-B4 fine-tuned on 50,000 image dataset. Achieves 82%+ accuracy on benchmark. Exported to ONNX.	ML Engineer
3	Upload API	POST /scan endpoint accepts image file up to 20MB. Returns job ID. Stores file in S3. Writes to PostgreSQL.	Backend Engineer
3	Job queue	Celery + Redis queue processes scan jobs. Workers pull jobs, run detection pipeline, write results. Handles 50 concurrent jobs.	Backend Engineer
4	Frontend upload UI	Next.js app. Drag and drop upload. Progress indicator. Works on mobile and desktop. Passes Lighthouse score 90+.	Frontend Engineer
4	Results page	Shows verdict, confidence score, and 3-signal breakdown (FFT, metadata, CNN). Mobile responsive.	Frontend Engineer
4	User accounts	Register, login, logout, password reset. JWT auth. Scan history page showing last 50 scans.	Backend Engineer
5	Basic API	Developer API key. POST /v1/scan endpoint. JSON response with verdict and signal scores. Rate limited by plan.	Backend Engineer
5	Public launch	Product live at verifai.io. White paper published. Press release to AI/media tech journalists. 50 beta users.	Founder
5	Monitoring	Prometheus + Grafana dashboards live. Sentry error tracking. PagerDuty alerts for downtime.	DevOps

Phase 2 — Precision (Months 6-9)

Goal

Push image detection accuracy to 92%+. Add the full signal stack. Get first paying customers. Get first university validation.

Month	Deliverable	Definition of Done	Owner
6	Vision Transformer (ViT)	ViT-L/16 fine-tuned and running alongside CNN. Ensemble output 89%+ accuracy. ONNX exported.	ML Engineer

6	Noise fingerprinting module	FAISS index of 10 generator fingerprints. Module identifies generator family with 78%+ accuracy.	ML Engineer
7	Facial analysis module	InsightFace pipeline. Detects eye asymmetry, skin texture anomaly, hand geometry. Adds 3% to ensemble accuracy.	ML Engineer
7	Ensemble scoring v2	Dynamic weight system. Weights adjust based on file type, resolution, face presence. Ensemble hits 91%+.	ML Engineer
7	Heatmap output	GradCAM heatmap generated for every scan. Overlaid on image. Downloadable as PNG.	ML + Frontend
8	Full explainability engine	Every result includes plain-English paragraph explaining why verdict was reached. Reviewed by legal consultant.	Backend + Founder
8	Stripe billing integration	Free, Pro (\$29), Business (\$99) plans. Upgrade/downgrade. Invoice history. Stripe webhook for subscription events.	Backend Engineer
8	University outreach	Formal research partnership agreed with 1 university. Data sharing agreement signed. Validation study timeline agreed.	Founder
9	Accuracy 92% milestone	Full benchmark run confirms 92%+ accuracy. Published on website with full methodology. Press release.	All
9	First 100 paying customers	100 Pro or Business plan subscribers. MRR \$3,000+. Collect and publish 10 case studies.	Founder

Phase 3 — Video (Months 10-14)

Goal

Full video detection capability. Handle files up to 2GB. 88%+ video accuracy. Raise seed funding during this phase.

Month	Deliverable	Definition of Done	Owner
10	Video frame pipeline	Adaptive keyframe sampler: scene detection + optical flow. Processes 60s video in under 30s. Unit tested.	ML Engineer
10	GPU-accelerated decode	NVDEC integration via PyAV. 8x faster than CPU decode. Benchmark confirmed.	ML Engineer

11	Temporal analysis module	Flickering detector, optical flow anomaly, object permanence tracker. Adds temporal signal to ensemble.	ML Engineer
11	Chunked upload	Uppy.js frontend chunking. Backend reassembles chunks. Supports files up to 2GB. Resumes after network drop.	Frontend + Backend
12	Video ensemble	Temporal signal weighted 25% in video ensemble. Video benchmark suite created. 88%+ accuracy confirmed.	ML Engineer
12	Video results UI	Timeline heatmap showing per-second confidence. Worst-offending frame highlighted. Download still frame.	Frontend Engineer
13	Seed funding raise	Pitch deck finalised. 20 investor meetings. Term sheet signed. \$1-2M raised at \$6-10M valuation.	Founder
13	Retraining pipeline v1	Airflow DAG orchestrates full retrain cycle. Runs end to end without manual intervention. Tested twice.	ML + DevOps
14	Early warning system	Bot monitors AI model releases. Auto-generates test images. Alerts team if detection drops on new model.	ML Engineer
14	500 paying customers	MRR \$15,000+. First enterprise pilot signed. API customers generating \$2,000+/month.	Founder

Phase 4 — Legal Grade (Months 15-18)

Goal

Court-admissible forensic reports. NIST evaluation submitted. Expert witness network. First legal case uses VerifAI evidence.

Month	Deliverable	Definition of Done	Owner
15	Chain of custody system	SHA-256 hash at upload. Tamper-evident audit log. Write-once S3 bucket. Tested by forensic consultant.	Backend + DevOps
15	Forensic report generator	Signed PDF report with all required legal fields. PKI certificate. Reviewed by 2 lawyers in 2 jurisdictions.	Backend + Founder
16	NIST evaluation submission	Full methodology document submitted to NIST FATE or PFT programme. Confirmation of receipt.	Founder + ML

16	University validation published	Partner university publishes independent accuracy study. VerifAI scores 91%+ in independent test.	University + Founder
17	Expert witness network	3 certified forensic analysts trained on VerifAI methodology. At least 1 has prior court expert witness experience.	Founder
17	White paper v2 published	Peer-reviewed paper submitted to major AI conference (NeurIPS, CVPR, or FAccT). Pre-print on arXiv.	ML Engineer
18	First legal case	VerifAI forensic report submitted as evidence in at least 1 real legal case. Outcome documented.	Founder + Analyst
18	2,000 paying customers	MRR \$65,000+. 2 enterprise contracts. Series A pitch ready.	Founder

Phase 5 — Standard (Months 19-24)

Goal

Become the reference standard. Platform API partnerships. Series A closed. 5,000+ paying customers. Referenced in legislation.

Month	Deliverable	Definition of Done	Owner
19	Series A close	\$8-15M raised. 3 strategic investors (media company or agency). Hire 6 additional team members.	Founder
19	C2PA integration	Content Credentials Verify integration complete. VerifAI checks C2PA signatures and reports on them.	Backend Engineer
20	Browser extension	Chrome extension lets users right-click any image to scan it. 10,000 installs in first month.	Frontend Engineer
20	Platform API v2	White-label API. Custom branding. Webhook events. SLA 99.9%. First 3 platform partners signed.	Backend + Founder
21	Mobile apps	iOS and Android apps. Camera roll scan, share sheet extension. 50,000 downloads in first month.	Mobile Engineer
22	50 generator fingerprints	Fingerprint database covers 50 generator versions. Attribution accuracy 90%+. Published generator list.	ML Engineer
22	Government partnerships	At least 1 government agency (election body, court system, or press regulator) using VerifAI officially.	Founder

23	94%+ image accuracy	Full benchmark confirms 94%+ image, 91%+ video. Published. Cited in academic papers.	ML Team
24	Industry standard status	Referenced in legislation or official guidance in at least 1 jurisdiction. Featured in AP, Reuters, BBC coverage.	Founder
24	5,000 paying customers	MRR \$350,000+. ARR \$4.2M+. EBITDA positive. Company valued at \$30-50M.	All

2. Full Cost Breakdown — Development & Production

2.1 Development Costs (One-Time, Phases 0-2)

Category	Item	Cost Estimate	Notes
Team	ML Engineer (12 months)	\$72,000	Can be equity-heavy early on
Team	Backend Engineer (12 months)	\$65,000	
Team	Frontend Engineer (12 months)	\$60,000	
Team	DevOps (part-time, 12 months)	\$30,000	Contractor initially
Team	Founder (no salary Year 1)	\$0	Defer until funded
Infrastructure	AWS (dev + staging)	\$4,800	\$400/month x 12
Infrastructure	GPU training runs (A100 spot)	\$3,600	\$300/month x 12
Data	Dataset annotation (human labelers)	\$12,000	Label Studio + contractors
Data	Real photo licenses (Flickr, stock)	\$2,400	\$200/month
Legal	Company formation + IP	\$3,000	
Legal	Privacy policy, T&Cs, GDPR review	\$4,000	
Tools	Dev tools, SaaS (GitHub, Sentry, etc.)	\$3,600	\$300/month
Research	Academic paper access, conferences	\$2,000	
Marketing	Website, domain, early content	\$3,000	
	TOTAL DEVELOPMENT COST (Year 1)	\$265,400	

2.2 Production Running Costs (Monthly, by Phase)

Cost Category	Phase 1 (Month 6)	Phase 2 (Month 12)	Phase 3 (Month 18)	Phase 4 (Month 24)
GPU inference (AWS g5)	\$800	\$2,400	\$6,000	\$14,000

GPU training (spot A100)	\$300	\$600	\$1,200	\$2,400
S3 storage + CDN	\$120	\$320	\$800	\$2,000
RDS PostgreSQL	\$150	\$300	\$600	\$1,200
Redis (ElastiCache)	\$80	\$160	\$320	\$640
Lambda + SQS	\$40	\$120	\$300	\$600
Monitoring (Grafana Cloud)	\$50	\$100	\$200	\$400
Team salaries	\$18,000	\$28,000	\$55,000	\$120,000
Data annotation	\$1,000	\$2,000	\$4,000	\$8,000
Legal + compliance	\$500	\$1,500	\$3,000	\$5,000
Marketing	\$500	\$2,000	\$8,000	\$20,000
Misc + buffer	\$500	\$1,000	\$2,000	\$4,000
TOTAL MONTHLY OPEX	\$22,040	\$38,500	\$81,420	\$178,240

2.3 Where to Cut Costs Without Cutting Quality

Biggest saving: GPU spot instances

AWS spot instances for training runs cost 70% less than on-demand. Training jobs can be interrupted and resumed, so spot is fine for training. Only use on-demand for inference (which must never be interrupted). Expected saving: \$800-2,000/month at scale.

Second biggest saving: ONNX inference optimisation

Converting PyTorch models to ONNX Runtime format gives 3-4x faster inference. This means you need fewer GPU instances to handle the same scan volume. At 10,000 scans/day, ONNX means 2 GPU instances instead of 6. Saving: \$3,200/month at scale.

Third biggest saving: cache deduplication

SHA-256 deduplication means the same file is never processed twice. At scale, 40-60% of all scans are duplicates (same viral image uploaded by many users). Each cache hit costs \$0.001 (Redis lookup) instead of \$0.08 (GPU inference). Saving at 100,000 scans/day: \$3,200/day.

Other cost reduction strategies

- Use Lambda for lightweight preprocessing tasks instead of always-on EC2 — only pay when a scan comes in

- CloudFront CDN serves the heatmap images and result assets — no repeated S3 reads at cost
- Use Hetzner or Vast.ai for training runs instead of AWS — GPU compute is 60% cheaper on these platforms
- Hire engineers from markets with strong talent and lower salary expectations (Nigeria, Eastern Europe, India) — same quality, 40-60% lower cost
- Open-source non-core components (the benchmark test suite) — community contributes improvements for free
- Use Cloudflare Pages for the frontend — free tier handles millions of page views
- DVC + S3 for dataset storage instead of a managed ML platform — saves \$2,000+/month vs Weights & Biases at scale

Cost Lever	Monthly Saving (Year 2)	Effort to Implement	Risk
Spot instances for training	\$1,200	Low — config change only	Minimal
ONNX inference	\$2,400	Medium — model conversion + testing	Low
Cache deduplication	\$4,800	Medium — already in architecture	None
Hetzner for training	\$800	Low — script change	Minimal
Cloudflare Pages frontend	\$200	Low	None
DVC instead of W&B	\$1,800	Medium	Low
Total monthly savings	\$11,200		

3. Full Financial Model

3.1 Revenue Build-Up (Month by Month, Year 1-2)

Month	Free Users	Pro (\$29)	Business (\$99)	API Rev	Reports	MRR
M3 (launch)	120	8	2	\$0	\$0	\$430
M4	280	18	5	\$200	\$0	\$1,217
M5	500	35	9	\$400	\$0	\$2,906
M6	900	60	15	\$800	\$300	\$4,125
M7	1,400	90	22	\$1,200	\$600	\$6,468
M8	2,200	130	32	\$2,000	\$900	\$9,678
M9	3,500	190	48	\$3,000	\$1,500	\$14,252
M10	5,000	270	68	\$4,500	\$2,000	\$20,232
M11	7,000	380	95	\$6,000	\$3,000	\$28,525
M12	10,000	520	130	\$8,000	\$5,000	\$39,380
M18	28,000	1,400	380	\$25,000	\$15,000	\$103,020
M24	60,000	3,200	900	\$60,000	\$40,000	\$262,600

3.2 Annual P&L Statement

	Year 1	Year 2	Year 3	Year 4	Year 5
REVENUE					
SaaS subscriptions	\$58k	\$620k	\$2.1M	\$5.8M	\$14.4M
API usage	\$18k	\$180k	\$600k	\$1.6M	\$4.0M
Forensic reports	\$8k	\$120k	\$500k	\$1.5M	\$4.0M
Enterprise / white-label	\$0	\$80k	\$500k	\$2.0M	\$6.0M
TOTAL REVENUE	\$84k	\$1.0M	\$3.7M	\$10.9M	\$28.4M
COSTS					
Team (salaries)	\$265k	\$620k	\$1.4M	\$2.8M	\$5.0M
Infrastructure (AWS)	\$18k	\$72k	\$240k	\$600k	\$1.2M
Data + annotation	\$14k	\$50k	\$120k	\$250k	\$400k

Legal + compliance	\$7k	\$40k	\$80k	\$150k	\$250k
Marketing + sales	\$5k	\$80k	\$300k	\$800k	\$2.0M
Research + partnerships	\$3k	\$30k	\$60k	\$100k	\$150k
G&A + misc	\$8k	\$30k	\$80k	\$200k	\$400k
TOTAL COSTS	\$320k	\$922k	\$2.28M	\$4.9M	\$9.4M
NET PROFIT / (LOSS)	(\$236k)	\$78k	\$1.42M	\$6.0M	\$19.0M
Net margin	-281%	8%	38%	55%	67%
Cumulative cash	(\$236k)	(\$158k)	\$1.26M	\$7.26M	\$26.26M

Note on Year 1 Loss

The \$236k loss in Year 1 is investment in the moat — the dataset, the retraining pipeline, the white paper. These are not wasted costs. They are the assets that justify the valuation and make every future year more profitable. This is normal for deep-tech products.

3.3 Unit Economics

Metric	Phase 1	Phase 2	At Scale (Year 3+)
Cost per image scan (GPU)	\$0.018	\$0.009	\$0.004
Cost per video scan (GPU)	\$0.12	\$0.06	\$0.025
Cost per scan after caching	\$0.011	\$0.005	\$0.002
Revenue per Pro user/month	\$29	\$29	\$29
Gross margin per Pro user	\$25.80 (89%)	\$26.50 (91%)	\$27.20 (94%)
Customer acquisition cost (CAC)	\$45	\$35	\$25
Lifetime value Pro user (LTV)	\$290 (10mo avg)	\$580 (20mo avg)	\$870 (30mo avg)
LTV:CAC ratio	6.4x	16.6x	34.8x
Payback period	1.5 months	1.2 months	0.9 months

4. Company Valuation and Exit Strategy

4.1 How SaaS Companies Are Valued

SaaS companies are typically valued on a multiple of their Annual Recurring Revenue (ARR). The multiple depends on growth rate, gross margin, market size, and defensibility of the business. AI infrastructure companies with strong moats command higher multiples than generic SaaS.

Valuation Method	Year 1	Year 2	Year 3	Year 4	Year 5
ARR	\$84k	\$1.0M	\$3.7M	\$10.9M	\$28.4M
Revenue multiple (conservative 8x)	\$672k	\$8M	\$29.6M	\$87.2M	\$227M
Revenue multiple (market 15x)	\$1.26M	\$15M	\$55.5M	\$163.5M	\$426M
Revenue multiple (premium 25x for standard)	N/A	\$25M	\$92.5M	\$272.5M	\$710M
DCF valuation (10% discount)	N/A	\$4M	\$28M	\$95M	\$220M
Estimated fair value range	\$2-5M	\$8-25M	\$30-90M	\$90-270M	\$220-700M

Why We Command a Premium Multiple

Generic SaaS gets 8-10x ARR. VerifAI commands 15-25x because: (1) we have a proprietary dataset that is a moat no one can replicate quickly, (2) we are positioning as infrastructure — the detection layer inside other platforms, (3) legal and government adoption creates switching costs and barriers to competition, (4) the market is growing at 42% CAGR and we have first-mover advantage on the legal-grade segment.

4.2 Funding Rounds and Dilution

Round	Timing	Amount	Valuation	Dilution	Use of Funds
Bootstrap	Month 1	\$50-150k	N/A	0%	Founder capital. Dev environment, dataset, first hire.
Pre-seed grant	Month 2-4	\$50-200k	N/A	0%	Google for Startups, Tony Elumelu, NIST SBIR. Non-dilutive.
Seed	Month 13-14	\$1-2M	\$6-10M pre	12-18%	Video pipeline, retraining system, first 3 enterprise sales hires.
Series A	Month 19-20	\$8-15M	\$40-60M pre	15-22%	Team scale to 20, platform API partnerships, mobile apps, legal track.

Series B (optional)	Month 30-36	\$30-50M	\$150-250M pre	15-20%	International expansion, government contracts, acquisition of dataset companies.
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Founder dilution after Series A (assuming 2 founders, 10% option pool): approximately 55-65% of the company retained. At a \$40-60M valuation, founder stake is worth \$22-39M on paper.

4.3 What You Need to Raise at the Idea Stage

Raising at idea stage (before a product) is hard but possible if you can demonstrate technical credibility and a clear path to the standard. Here is exactly what investors need to see:

The 6 things that make idea-stage investors say yes

- A technical founder who clearly understands the ML deeply — not someone who just had the idea
- A published benchmark methodology (even a draft white paper shows you know what you are building)
- Evidence of the market problem — real examples of AI-generated content causing harm (legal cases, fraud incidents, misinformation examples)
- A specific path to defensibility — explain the proprietary dataset strategy and the retraining pipeline
- A realistic go-to-market — not 'we will sell to everyone' but 'we start with investigative journalists and legal firms, who have the most urgent need and will pay premium prices'
- Founder credibility — relevant background in ML, computer vision, forensics, or media tech

What to prepare for investor meetings

- One-pager: 1 page, problem/solution/market/traction/ask. Can be read in 90 seconds.
- Pitch deck: 12-15 slides. Problem, solution, how it works (technical but clear), market size, competition, business model, traction, team, roadmap, ask.
- Technical demo: even a prototype showing FFT analysis on a sample image proves you can build it
- White paper draft: shows scientific credibility. Does not need to be published yet.
- Financial model: 3-year projections. Investors will stress-test the assumptions.
- Data room: company docs, IP assignment, cap table, any LOIs from potential customers

4.4 Exit Scenarios — What We Could Sell For

Exit Scenario	Timing	Acquirer Type	Valuation Range	Rationale
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Early acquisition	Year 2-3	Adobe, Getty Images, Shutterstock	\$15-50M	They need detection for their own platforms. Buy vs build decision.
Strategic acquisition	Year 3-4	Microsoft, Google, Amazon	\$80-200M	AI infrastructure layer for their cloud offerings.
Media/news acquisition	Year 3-5	AP, Reuters, BBC, Axel Springer	\$60-150M	They need the standard. VerifAI becomes their owned standard.
Government/defence	Year 4-6	Palantir, BAE Systems, L3Harris	\$150-400M	National security use case for deepfake detection.
Private equity roll-up	Year 4-6	PE firms building forensics platforms	\$200-500M	Buy profitable SaaS with high margins and recurring revenue.
IPO	Year 6-8	Public markets	\$500M-2B+	If we achieve true industry standard status and \$50M+ ARR.

Most Likely Exit

A strategic acquisition by a major tech or media company in Year 3-5 at \$80-250M is the most realistic outcome. The IPO path requires achieving true standard status (referenced in legislation, used by governments) and \$50M+ ARR, which is achievable by Year 6-7 if we execute well.

5. Backend Architecture — Supabase / Convex Consideration

You asked whether VerifAI could be built using Supabase or Convex as the backend. This is a smart question about choosing the right tool for each part of the system. The answer is: yes for parts of it, with important caveats.

5.1 What Supabase Gives You

- PostgreSQL database — fully managed, with row-level security, real-time subscriptions, and pgvector extension
- Auth system — email/password, OAuth, magic links out of the box
- Storage — S3-compatible file storage
- Edge functions — Deno-based serverless functions
- Real-time — WebSocket subscriptions to database changes

Supabase Verdict for VerifAI

Supabase is an excellent choice for the database, auth, and file storage layers — especially in Phases 0-2. The PostgreSQL it provides is production-grade. The pgvector extension is available and is exactly what we need for the generator fingerprint database. Auth and storage are handled without building from scratch. Use Supabase for the data layer. It saves 3-4 months of backend infrastructure work.

5.2 What Convex Gives You

- A real-time document database with TypeScript-native queries
- Server functions that run in a transactional context
- Built-in file storage
- Real-time reactivity — UI updates automatically when data changes

Convex Verdict for VerifAI

Convex is excellent for real-time consumer apps but is less suited to VerifAI's needs. We have complex relational data (users, scans, signals, reports all linked), need pgvector for similarity search, and need Celery-style background job processing. Convex does not handle these well. It would be a constraint, not an enabler. Use Supabase over Convex for this project.

5.3 Recommended Hybrid Architecture (Supabase + FastAPI)

Layer	Tool	Justification
Database	Supabase PostgreSQL	Managed, production-grade, pgvector included, row-level security

Auth	Supabase Auth	Saves 2-3 weeks of development. JWT compatible with FastAPI.
File storage	Supabase Storage (Phase 1) → AWS S3 (Phase 2+)	Start simple, migrate to S3 when you need write-once buckets for forensic use
API layer	FastAPI (Python)	Required for ML pipeline. Cannot run PyTorch in Supabase edge functions.
Background jobs	Celery + Redis	Required for async scan processing. Supabase has no native job queue.
ML inference	FastAPI workers on AWS GPU	Cannot be replaced by Supabase — must be custom Python
Real-time status	Supabase real-time subscriptions	Client subscribes to scan status changes — no polling needed
Frontend	Next.js + Supabase JS client	Supabase client handles auth and data queries cleanly

In plain terms: use Supabase to eliminate all the boring database and auth infrastructure work, so your team can focus entirely on the ML pipeline — which is where your competitive advantage actually lives. Then as you scale and need forensic-grade storage and more control, migrate the storage layer to pure AWS S3 while keeping Supabase for the database.

5.4 What You Need to Learn

You said you are willing to learn everything needed. Here is the honest learning path, in priority order:

Skill	Priority	Time to Useful	Resources
Python (core)	Critical	4-6 weeks	Python for Everybody (Coursera), then build small scripts daily
PyTorch basics	Critical	8-12 weeks	fast.ai course, then PyTorch official tutorials
FastAPI	Critical	2-3 weeks	Official FastAPI docs are excellent. Build a simple API first.
PostgreSQL + SQL	Critical	3-4 weeks	Mode Analytics SQL Tutorial, then practice on real data
Supabase	High	1-2 weeks	Supabase official docs and YouTube channel
Next.js + React	High	6-8 weeks	Official Next.js tutorial, then build the upload UI
Docker basics	High	2-3 weeks	Docker official get-started guide
Computer vision basics	High	12-16 weeks	CS231n (Stanford, free online), OpenCV tutorials
AWS fundamentals	Medium	4-6 weeks	AWS Cloud Practitioner course (free tier)

MLflow + DVC	Medium	2-3 weeks	Official docs for both — very well documented
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Honest Advice

You do not need to master all of this before starting. The order that matters: Python first, then FastAPI to build the API, then Supabase to store data, then PyTorch to run a first model. You can have a working prototype in 3-4 months of focused learning + building. The rest of the skills come as you build. Learning by building is always faster than learning then building.

VERIFAI

Full Development Roadmap, Cost Breakdown & Financial Model

Version 3.0 | June 2026 | Confidential

1. Full Development Roadmap (Month by Month)

This roadmap covers 24 months from zero to industry standard. Each month has specific deliverables, milestones, and the person or team responsible. Nothing is vague — every month has a clear definition of done.

Phase 0 — Pre-Build (Months 1-2)

Goal

Have the scientific foundation, dataset pipeline, and development environment fully set up before writing a single line of product code. Most startups skip this phase and pay for it later. We do not.

Month	Deliverable	Definition of Done	Owner
1	Research review	Read and summarise top 20 academic papers on AI image detection. Extract key signal methods we will use.	ML Engineer
1	Dataset pipeline v1	Automated script collects 10,000 real images (Flickr API, Unsplash API) and generates 10,000 AI images across 5 generators.	ML Engineer
1	Dev environment	Docker Compose setup running FastAPI + PostgreSQL + Redis locally. All engineers can run the stack in one command.	Backend Engineer
1	Benchmark test suite	Python script that takes a model and outputs accuracy, FP rate, FN rate across all generator families. Will be used every sprint.	ML Engineer
2	Frequency analysis module	Standalone Python module: input image, output FFT features + anomaly score. Unit tested. Accuracy 72%+ on test set.	ML Engineer
2	Metadata extraction module	Standalone module: input file, output 8 metadata signals + rule-based score. Works on JPG, PNG, WEBP.	Backend Engineer
2	White paper draft v1	10-page technical paper describing our methodology. Not published yet — internal review only.	ML Engineer + Founder
2	AWS account + Terraform	All infrastructure defined as code. One command deploys a full staging environment.	DevOps

Phase 1 — MVP (Months 3-5)

Goal

A working product that real users can use. Image detection only. Target: 85% accuracy. Launch publicly at end of Month 5.

Month	Deliverable	Definition of Done	Owner
3	CNN classifier v1	EfficientNet-B4 fine-tuned on 50,000 image dataset. Achieves 82%+ accuracy on benchmark. Exported to ONNX.	ML Engineer
3	Upload API	POST /scan endpoint accepts image file up to 20MB. Returns job ID. Stores file in S3. Writes to PostgreSQL.	Backend Engineer
3	Job queue	Celery + Redis queue processes scan jobs. Workers pull jobs, run detection pipeline, write results. Handles 50 concurrent jobs.	Backend Engineer
4	Frontend upload UI	Next.js app. Drag and drop upload. Progress indicator. Works on mobile and desktop. Passes Lighthouse score 90+.	Frontend Engineer
4	Results page	Shows verdict, confidence score, and 3-signal breakdown (FFT, metadata, CNN). Mobile responsive.	Frontend Engineer
4	User accounts	Register, login, logout, password reset. JWT auth. Scan history page showing last 50 scans.	Backend Engineer
5	Basic API	Developer API key. POST /v1/scan endpoint. JSON response with verdict and signal scores. Rate limited by plan.	Backend Engineer
5	Public launch	Product live at verifai.io. White paper published. Press release to AI/media tech journalists. 50 beta users.	Founder
5	Monitoring	Prometheus + Grafana dashboards live. Sentry error tracking. PagerDuty alerts for downtime.	DevOps

Phase 2 — Precision (Months 6-9)

Goal

Push image detection accuracy to 92%+. Add the full signal stack. Get first paying customers. Get first university validation.

Month	Deliverable	Definition of Done	Owner
6	Vision Transformer (ViT)	ViT-L/16 fine-tuned and running alongside CNN. Ensemble output 89%+ accuracy. ONNX exported.	ML Engineer

6	Noise fingerprinting module	FAISS index of 10 generator fingerprints. Module identifies generator family with 78%+ accuracy.	ML Engineer
7	Facial analysis module	InsightFace pipeline. Detects eye asymmetry, skin texture anomaly, hand geometry. Adds 3% to ensemble accuracy.	ML Engineer
7	Ensemble scoring v2	Dynamic weight system. Weights adjust based on file type, resolution, face presence. Ensemble hits 91%+.	ML Engineer
7	Heatmap output	GradCAM heatmap generated for every scan. Overlaid on image. Downloadable as PNG.	ML + Frontend
8	Full explainability engine	Every result includes plain-English paragraph explaining why verdict was reached. Reviewed by legal consultant.	Backend + Founder
8	Stripe billing integration	Free, Pro (\$29), Business (\$99) plans. Upgrade/downgrade. Invoice history. Stripe webhook for subscription events.	Backend Engineer
8	University outreach	Formal research partnership agreed with 1 university. Data sharing agreement signed. Validation study timeline agreed.	Founder
9	Accuracy 92% milestone	Full benchmark run confirms 92%+ accuracy. Published on website with full methodology. Press release.	All
9	First 100 paying customers	100 Pro or Business plan subscribers. MRR \$3,000+. Collect and publish 10 case studies.	Founder

Phase 3 — Video (Months 10-14)

Goal

Full video detection capability. Handle files up to 2GB. 88%+ video accuracy. Raise seed funding during this phase.

Month	Deliverable	Definition of Done	Owner
10	Video frame pipeline	Adaptive keyframe sampler: scene detection + optical flow. Processes 60s video in under 30s. Unit tested.	ML Engineer
10	GPU-accelerated decode	NVDEC integration via PyAV. 8x faster than CPU decode. Benchmark confirmed.	ML Engineer

11	Temporal analysis module	Flickering detector, optical flow anomaly, object permanence tracker. Adds temporal signal to ensemble.	ML Engineer
11	Chunked upload	Uppy.js frontend chunking. Backend reassembles chunks. Supports files up to 2GB. Resumes after network drop.	Frontend + Backend
12	Video ensemble	Temporal signal weighted 25% in video ensemble. Video benchmark suite created. 88%+ accuracy confirmed.	ML Engineer
12	Video results UI	Timeline heatmap showing per-second confidence. Worst-offending frame highlighted. Download still frame.	Frontend Engineer
13	Seed funding raise	Pitch deck finalised. 20 investor meetings. Term sheet signed. \$1-2M raised at \$6-10M valuation.	Founder
13	Retraining pipeline v1	Airflow DAG orchestrates full retrain cycle. Runs end to end without manual intervention. Tested twice.	ML + DevOps
14	Early warning system	Bot monitors AI model releases. Auto-generates test images. Alerts team if detection drops on new model.	ML Engineer
14	500 paying customers	MRR \$15,000+. First enterprise pilot signed. API customers generating \$2,000+/month.	Founder

Phase 4 — Legal Grade (Months 15-18)

Goal

Court-admissible forensic reports. NIST evaluation submitted. Expert witness network. First legal case uses VerifAI evidence.

Month	Deliverable	Definition of Done	Owner
15	Chain of custody system	SHA-256 hash at upload. Tamper-evident audit log. Write-once S3 bucket. Tested by forensic consultant.	Backend + DevOps
15	Forensic report generator	Signed PDF report with all required legal fields. PKI certificate. Reviewed by 2 lawyers in 2 jurisdictions.	Backend + Founder
16	NIST evaluation submission	Full methodology document submitted to NIST FATE or PFT programme. Confirmation of receipt.	Founder + ML

16	University validation published	Partner university publishes independent accuracy study. VerifAI scores 91%+ in independent test.	University + Founder
17	Expert witness network	3 certified forensic analysts trained on VerifAI methodology. At least 1 has prior court expert witness experience.	Founder
17	White paper v2 published	Peer-reviewed paper submitted to major AI conference (NeurIPS, CVPR, or FAccT). Pre-print on arXiv.	ML Engineer
18	First legal case	VerifAI forensic report submitted as evidence in at least 1 real legal case. Outcome documented.	Founder + Analyst
18	2,000 paying customers	MRR \$65,000+. 2 enterprise contracts. Series A pitch ready.	Founder

Phase 5 — Standard (Months 19-24)

Goal

Become the reference standard. Platform API partnerships. Series A closed. 5,000+ paying customers. Referenced in legislation.

Month	Deliverable	Definition of Done	Owner
19	Series A close	\$8-15M raised. 3 strategic investors (media company or agency). Hire 6 additional team members.	Founder
19	C2PA integration	Content Credentials Verify integration complete. VerifAI checks C2PA signatures and reports on them.	Backend Engineer
20	Browser extension	Chrome extension lets users right-click any image to scan it. 10,000 installs in first month.	Frontend Engineer
20	Platform API v2	White-label API. Custom branding. Webhook events. SLA 99.9%. First 3 platform partners signed.	Backend + Founder
21	Mobile apps	iOS and Android apps. Camera roll scan, share sheet extension. 50,000 downloads in first month.	Mobile Engineer
22	50 generator fingerprints	Fingerprint database covers 50 generator versions. Attribution accuracy 90%+. Published generator list.	ML Engineer
22	Government partnerships	At least 1 government agency (election body, court system, or press regulator) using VerifAI officially.	Founder

23	94%+ image accuracy	Full benchmark confirms 94%+ image, 91%+ video. Published. Cited in academic papers.	ML Team
24	Industry standard status	Referenced in legislation or official guidance in at least 1 jurisdiction. Featured in AP, Reuters, BBC coverage.	Founder
24	5,000 paying customers	MRR \$350,000+. ARR \$4.2M+. EBITDA positive. Company valued at \$30-50M.	All

2. Full Cost Breakdown — Development & Production

2.1 Development Costs (One-Time, Phases 0-2)

Category	Item	Cost Estimate	Notes
Team	ML Engineer (12 months)	\$72,000	Can be equity-heavy early on
Team	Backend Engineer (12 months)	\$65,000	
Team	Frontend Engineer (12 months)	\$60,000	
Team	DevOps (part-time, 12 months)	\$30,000	Contractor initially
Team	Founder (no salary Year 1)	\$0	Defer until funded
Infrastructure	AWS (dev + staging)	\$4,800	\$400/month x 12
Infrastructure	GPU training runs (A100 spot)	\$3,600	\$300/month x 12
Data	Dataset annotation (human labelers)	\$12,000	Label Studio + contractors
Data	Real photo licenses (Flickr, stock)	\$2,400	\$200/month
Legal	Company formation + IP	\$3,000	
Legal	Privacy policy, T&Cs, GDPR review	\$4,000	
Tools	Dev tools, SaaS (GitHub, Sentry, etc.)	\$3,600	\$300/month
Research	Academic paper access, conferences	\$2,000	
Marketing	Website, domain, early content	\$3,000	
	TOTAL DEVELOPMENT COST (Year 1)	\$265,400	

2.2 Production Running Costs (Monthly, by Phase)

Cost Category	Phase 1 (Month 6)	Phase 2 (Month 12)	Phase 3 (Month 18)	Phase 4 (Month 24)
GPU inference (AWS g5)	\$800	\$2,400	\$6,000	\$14,000

GPU training (spot A100)	\$300	\$600	\$1,200	\$2,400
S3 storage + CDN	\$120	\$320	\$800	\$2,000
RDS PostgreSQL	\$150	\$300	\$600	\$1,200
Redis (ElastiCache)	\$80	\$160	\$320	\$640
Lambda + SQS	\$40	\$120	\$300	\$600
Monitoring (Grafana Cloud)	\$50	\$100	\$200	\$400
Team salaries	\$18,000	\$28,000	\$55,000	\$120,000
Data annotation	\$1,000	\$2,000	\$4,000	\$8,000
Legal + compliance	\$500	\$1,500	\$3,000	\$5,000
Marketing	\$500	\$2,000	\$8,000	\$20,000
Misc + buffer	\$500	\$1,000	\$2,000	\$4,000
TOTAL MONTHLY OPEX	\$22,040	\$38,500	\$81,420	\$178,240

2.3 Where to Cut Costs Without Cutting Quality

Biggest saving: GPU spot instances

AWS spot instances for training runs cost 70% less than on-demand. Training jobs can be interrupted and resumed, so spot is fine for training. Only use on-demand for inference (which must never be interrupted). Expected saving: \$800-2,000/month at scale.

Second biggest saving: ONNX inference optimisation

Converting PyTorch models to ONNX Runtime format gives 3-4x faster inference. This means you need fewer GPU instances to handle the same scan volume. At 10,000 scans/day, ONNX means 2 GPU instances instead of 6. Saving: \$3,200/month at scale.

Third biggest saving: cache deduplication

SHA-256 deduplication means the same file is never processed twice. At scale, 40-60% of all scans are duplicates (same viral image uploaded by many users). Each cache hit costs \$0.001 (Redis lookup) instead of \$0.08 (GPU inference). Saving at 100,000 scans/day: \$3,200/day.

Other cost reduction strategies

- Use Lambda for lightweight preprocessing tasks instead of always-on EC2 — only pay when a scan comes in

- CloudFront CDN serves the heatmap images and result assets — no repeated S3 reads at cost
- Use Hetzner or Vast.ai for training runs instead of AWS — GPU compute is 60% cheaper on these platforms
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REVENUE					
SaaS subscriptions	\$58k	\$620k	\$2.1M	\$5.8M	\$14.4M
API usage	\$18k	\$180k	\$600k	\$1.6M	\$4.0M
Forensic reports	\$8k	\$120k	\$500k	\$1.5M	\$4.0M
Enterprise / white-label	\$0	\$80k	\$500k	\$2.0M	\$6.0M
TOTAL REVENUE	\$84k	\$1.0M	\$3.7M	\$10.9M	\$28.4M
COSTS					
Team (salaries)	\$265k	\$620k	\$1.4M	\$2.8M	\$5.0M
Infrastructure (AWS)	\$18k	\$72k	\$240k	\$600k	\$1.2M
Data + annotation	\$14k	\$50k	\$120k	\$250k	\$400k

Legal + compliance	\$7k	\$40k	\$80k	\$150k	\$250k
Marketing + sales	\$5k	\$80k	\$300k	\$800k	\$2.0M
Research + partnerships	\$3k	\$30k	\$60k	\$100k	\$150k
G&A + misc	\$8k	\$30k	\$80k	\$200k	\$400k
TOTAL COSTS	\$320k	\$922k	\$2.28M	\$4.9M	\$9.4M
NET PROFIT / (LOSS)	(\$236k)	\$78k	\$1.42M	\$6.0M	\$19.0M
Net margin	-281%	8%	38%	55%	67%
Cumulative cash	(\$236k)	(\$158k)	\$1.26M	\$7.26M	\$26.26M

Note on Year 1 Loss

The \$236k loss in Year 1 is investment in the moat — the dataset, the retraining pipeline, the white paper. These are not wasted costs. They are the assets that justify the valuation and make every future year more profitable. This is normal for deep-tech products.

3.3 Unit Economics

Metric	Phase 1	Phase 2	At Scale (Year 3+)
Cost per image scan (GPU)	\$0.018	\$0.009	\$0.004
Cost per video scan (GPU)	\$0.12	\$0.06	\$0.025
Cost per scan after caching	\$0.011	\$0.005	\$0.002
Revenue per Pro user/month	\$29	\$29	\$29
Gross margin per Pro user	\$25.80 (89%)	\$26.50 (91%)	\$27.20 (94%)
Customer acquisition cost (CAC)	\$45	\$35	\$25
Lifetime value Pro user (LTV)	\$290 (10mo avg)	\$580 (20mo avg)	\$870 (30mo avg)
LTV:CAC ratio	6.4x	16.6x	34.8x
Payback period	1.5 months	1.2 months	0.9 months

4. Company Valuation and Exit Strategy

4.1 How SaaS Companies Are Valued

SaaS companies are typically valued on a multiple of their Annual Recurring Revenue (ARR). The multiple depends on growth rate, gross margin, market size, and defensibility of the business. AI infrastructure companies with strong moats command higher multiples than generic SaaS.

Valuation Method	Year 1	Year 2	Year 3	Year 4	Year 5
ARR	\$84k	\$1.0M	\$3.7M	\$10.9M	\$28.4M
Revenue multiple (conservative 8x)	\$672k	\$8M	\$29.6M	\$87.2M	\$227M
Revenue multiple (market 15x)	\$1.26M	\$15M	\$55.5M	\$163.5M	\$426M
Revenue multiple (premium 25x for standard)	N/A	\$25M	\$92.5M	\$272.5M	\$710M
DCF valuation (10% discount)	N/A	\$4M	\$28M	\$95M	\$220M
Estimated fair value range	\$2-5M	\$8-25M	\$30-90M	\$90-270M	\$220-700M

Why We Command a Premium Multiple

Generic SaaS gets 8-10x ARR. VerifAI commands 15-25x because: (1) we have a proprietary dataset that is a moat no one can replicate quickly, (2) we are positioning as infrastructure — the detection layer inside other platforms, (3) legal and government adoption creates switching costs and barriers to competition, (4) the market is growing at 42% CAGR and we have first-mover advantage on the legal-grade segment.

4.2 Funding Rounds and Dilution

Round	Timing	Amount	Valuation	Dilution	Use of Funds
Bootstrap	Month 1	\$50-150k	N/A	0%	Founder capital. Dev environment, dataset, first hire.
Pre-seed grant	Month 2-4	\$50-200k	N/A	0%	Google for Startups, Tony Elumelu, NIST SBIR. Non-dilutive.
Seed	Month 13-14	\$1-2M	\$6-10M pre	12-18%	Video pipeline, retraining system, first 3 enterprise sales hires.
Series A	Month 19-20	\$8-15M	\$40-60M pre	15-22%	Team scale to 20, platform API partnerships, mobile apps, legal track.

Series B (optional)	Month 30-36	\$30-50M	\$150-250M pre	15-20%	International expansion, government contracts, acquisition of dataset companies.
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Founder dilution after Series A (assuming 2 founders, 10% option pool): approximately 55-65% of the company retained. At a \$40-60M valuation, founder stake is worth \$22-39M on paper.

4.3 What You Need to Raise at the Idea Stage

Raising at idea stage (before a product) is hard but possible if you can demonstrate technical credibility and a clear path to the standard. Here is exactly what investors need to see:

The 6 things that make idea-stage investors say yes

- A technical founder who clearly understands the ML deeply — not someone who just had the idea
- A published benchmark methodology (even a draft white paper shows you know what you are building)
- Evidence of the market problem — real examples of AI-generated content causing harm (legal cases, fraud incidents, misinformation examples)
- A specific path to defensibility — explain the proprietary dataset strategy and the retraining pipeline
- A realistic go-to-market — not 'we will sell to everyone' but 'we start with investigative journalists and legal firms, who have the most urgent need and will pay premium prices'
- Founder credibility — relevant background in ML, computer vision, forensics, or media tech

What to prepare for investor meetings

- One-pager: 1 page, problem/solution/market/traction/ask. Can be read in 90 seconds.
- Pitch deck: 12-15 slides. Problem, solution, how it works (technical but clear), market size, competition, business model, traction, team, roadmap, ask.
- Technical demo: even a prototype showing FFT analysis on a sample image proves you can build it
- White paper draft: shows scientific credibility. Does not need to be published yet.
- Financial model: 3-year projections. Investors will stress-test the assumptions.
- Data room: company docs, IP assignment, cap table, any LOIs from potential customers

4.4 Exit Scenarios — What We Could Sell For

Exit Scenario	Timing	Acquirer Type	Valuation Range	Rationale
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Early acquisition	Year 2-3	Adobe, Getty Images, Shutterstock	\$15-50M	They need detection for their own platforms. Buy vs build decision.
Strategic acquisition	Year 3-4	Microsoft, Google, Amazon	\$80-200M	AI infrastructure layer for their cloud offerings.
Media/news acquisition	Year 3-5	AP, Reuters, BBC, Axel Springer	\$60-150M	They need the standard. VerifAI becomes their owned standard.
Government/defence	Year 4-6	Palantir, BAE Systems, L3Harris	\$150-400M	National security use case for deepfake detection.
Private equity roll-up	Year 4-6	PE firms building forensics platforms	\$200-500M	Buy profitable SaaS with high margins and recurring revenue.
IPO	Year 6-8	Public markets	\$500M-2B+	If we achieve true industry standard status and \$50M+ ARR.

Most Likely Exit

A strategic acquisition by a major tech or media company in Year 3-5 at \$80-250M is the most realistic outcome. The IPO path requires achieving true standard status (referenced in legislation, used by governments) and \$50M+ ARR, which is achievable by Year 6-7 if we execute well.

5. Backend Architecture — Supabase / Convex Consideration

You asked whether VerifAI could be built using Supabase or Convex as the backend. This is a smart question about choosing the right tool for each part of the system. The answer is: yes for parts of it, with important caveats.

5.1 What Supabase Gives You

- PostgreSQL database — fully managed, with row-level security, real-time subscriptions, and pgvector extension
- Auth system — email/password, OAuth, magic links out of the box
- Storage — S3-compatible file storage
- Edge functions — Deno-based serverless functions
- Real-time — WebSocket subscriptions to database changes

Supabase Verdict for VerifAI

Supabase is an excellent choice for the database, auth, and file storage layers — especially in Phases 0-2. The PostgreSQL it provides is production-grade. The pgvector extension is available and is exactly what we need for the generator fingerprint database. Auth and storage are handled without building from scratch. Use Supabase for the data layer. It saves 3-4 months of backend infrastructure work.

5.2 What Convex Gives You

- A real-time document database with TypeScript-native queries
- Server functions that run in a transactional context
- Built-in file storage
- Real-time reactivity — UI updates automatically when data changes

Convex Verdict for VerifAI

Convex is excellent for real-time consumer apps but is less suited to VerifAI's needs. We have complex relational data (users, scans, signals, reports all linked), need pgvector for similarity search, and need Celery-style background job processing. Convex does not handle these well. It would be a constraint, not an enabler. Use Supabase over Convex for this project.

5.3 Recommended Hybrid Architecture (Supabase + FastAPI)

Layer	Tool	Justification
Database	Supabase PostgreSQL	Managed, production-grade, pgvector included, row-level security

Auth	Supabase Auth	Saves 2-3 weeks of development. JWT compatible with FastAPI.
File storage	Supabase Storage (Phase 1) → AWS S3 (Phase 2+)	Start simple, migrate to S3 when you need write-once buckets for forensic use
API layer	FastAPI (Python)	Required for ML pipeline. Cannot run PyTorch in Supabase edge functions.
Background jobs	Celery + Redis	Required for async scan processing. Supabase has no native job queue.
ML inference	FastAPI workers on AWS GPU	Cannot be replaced by Supabase — must be custom Python
Real-time status	Supabase real-time subscriptions	Client subscribes to scan status changes — no polling needed
Frontend	Next.js + Supabase JS client	Supabase client handles auth and data queries cleanly

In plain terms: use Supabase to eliminate all the boring database and auth infrastructure work, so your team can focus entirely on the ML pipeline — which is where your competitive advantage actually lives. Then as you scale and need forensic-grade storage and more control, migrate the storage layer to pure AWS S3 while keeping Supabase for the database.

5.4 What You Need to Learn

You said you are willing to learn everything needed. Here is the honest learning path, in priority order:

Skill	Priority	Time to Useful	Resources
Python (core)	Critical	4-6 weeks	Python for Everybody (Coursera), then build small scripts daily
PyTorch basics	Critical	8-12 weeks	fast.ai course, then PyTorch official tutorials
FastAPI	Critical	2-3 weeks	Official FastAPI docs are excellent. Build a simple API first.
PostgreSQL + SQL	Critical	3-4 weeks	Mode Analytics SQL Tutorial, then practice on real data
Supabase	High	1-2 weeks	Supabase official docs and YouTube channel
Next.js + React	High	6-8 weeks	Official Next.js tutorial, then build the upload UI
Docker basics	High	2-3 weeks	Docker official get-started guide
Computer vision basics	High	12-16 weeks	CS231n (Stanford, free online), OpenCV tutorials
AWS fundamentals	Medium	4-6 weeks	AWS Cloud Practitioner course (free tier)

MLflow + DVC	Medium	2-3 weeks	Official docs for both — very well documented
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Honest Advice

You do not need to master all of this before starting. The order that matters: Python first, then FastAPI to build the API, then Supabase to store data, then PyTorch to run a first model. You can have a working prototype in 3-4 months of focused learning + building. The rest of the skills come as you build. Learning by building is always faster than learning then building.